Chapter 22

Covariates, Longitudinal Profiling, and Repeated Measures in Mass Spectrometry: Linear Mixed Models, Variance Partitioning, and Longitudinal Splines

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Abstract

Quantitative mass spectrometry (MS) experiments in clinical and translational cohorts routinely depart from simple, independent two-group cross-sectional designs. Modern biomolecular investigations frequently feature paired pre- and post-treatment biopsies, multi-day technical batch runs, complex factorial interventions, and multi-timepoint longitudinal monitoring of the same subjects. Applying standard Ordinary Least Squares (OLS) models, two-sample Student's t-tests, or standard unblocked ANOVA to such structured datasets violates the core assumption of independent and identically distributed (i.i.d.) observations. This analytical error—known as pseudoreplication—underestimates parameter standard errors, produces artificially deflated p-values, and inundates the literature with irreproducible biomarker candidates. Conversely, failing to adjust for technical confounders (such as LC-MS run day, operator, or autosampler tray) inflates residual variance and obscures genuine biological effects.

This chapter establishes a rigorous mathematical, methodological, and computational framework for modeling complex experimental structure in high-dimensional proteomics and metabolomics. We develop the matrix formulation of Linear Mixed-Effects Models (LMMs), provide an algebraic derivation of Henderson's mixed-model equations from joint log-densities, and contrast Restricted Maximum Likelihood (REML) against Maximum Likelihood (ML) estimation. We formalize the Intraclass Correlation Coefficient (ICC) and provide an analytical proof of the variance inflation factor induced by ignoring repeated measures. We evaluate alternative covariance structures—including Compound Symmetry (CS), First-Order Autoregressive ($\mathrm{AR}(1)$*), and Random Coefficient models—and establish natural cubic splines for capturing non-linear kinetic time courses. Furthermore, we bridge linear mixed models with empirical Bayes moderation and variance partitioning through the* `dream` *and* `variancePartition` *frameworks. The chapter features a tutorial diagnostic heuristics checklist, an end-to-end reproducible R laboratory on a published targeted mass spectrometry longitudinal time-course, a critical review of the "Five Deadly Sins" of longitudinal MS analysis, and multi-tier exercises with exhaustive, step-by-step worked solutions.*

Keywords: Linear Mixed-Effects Models, Repeated Measures, Longitudinal Profiling, Intraclass Correlation Coefficient (ICC), REML, Pseudoreplication, Splines, Empirical Bayes Moderation, `dream`, `variancePartition`,

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Kenward-Roger.

22.1 Introduction: The Structural Complexity of Omics Experiments

A pervasive misconception in high-throughput biomolecular analysis is that differential abundance analysis is fundamentally an exercise in two-group hypothesis testing. In clinical proteomics, drug discovery, and biomarker validation, samples are almost never purely independent draws from an exchangeable population [3]. Real-world mass spectrometry studies are characterized by multiple layers of hierarchy, grouping, and repeated sampling:

- **Longitudinal Time Courses:** In pharmacodynamic and vaccination studies, biospecimens (e.g., plasma, serum, tissue biopsies) are collected from the same individual patient across multiple consecutive timepoints (e.g., baseline, 2 hours, 24 hours, 7 days post-treatment) [11].
- **Matched and Paired Cohorts:** Studies often utilize paired designs where diseased and adjacent non-diseased tissue are resected from the same cancer patient, or where subjects serve as their own internal controls in cross-over trials.
- **Hierarchical / Nested Sampling:** High-content single-cell or spatial MS workflows measure multiple single cells, laser-capture microdissection (LCM) spots, or technical instrument replicates nested within specific tissue blocks, patients, or microtiter plates [12].
- **Extraneous Technical Confounders:** Samples analyzed over weeks or months inevitably span multiple LC columns, chromatography gradients, instrument calibrations, digestion batches, and operator shifts [25, 21, 5].

When structured or longitudinal observations are analyzed with naive methods that treat each mass spectrum as an independent realization, standard errors are artificially compressed by a factor that depends on the magnitude of the within-subject correlation. This structural mismatch produces severe false-positive inflation. Conversely, when large patient-to-patient baseline variability is left unmodeled in paired designs, genuine drug responses are swallowed by inter-individual heterogeneity, producing false negatives [4]. The statistical model must faithfully reflect the physical architecture of the experiment [8, 9, 7, 19].

22.2 Mathematical Foundations of Linear Mixed-Effects Models

22.2.1 Matrix Formulation and Hierarchical Structure

To accommodate both population-level trends (fixed effects) and subject-specific deviations (random effects), we formalize the Linear Mixed-Effects Model (LMM). Let N denote the total number of subjects (clusters), with subject $i \in \{1,\dots,N\}$ measured at n_i occasions, yielding a total sample size of $n_{\text{total}} = \sum_{i=1}^{N} n_i$. For a given mass spectrometry feature g (peptide, protein, or metabolite), let $\mathbf{y}_i \in \mathbb{R}^{n_i}$ denote the vector of observed $\log_2$-transformed abundance values for subject i. The general mixed-effects model is formulated as [28, 24]:

$$\mathbf{y}_i = \mathbf{X}_i\beta + \mathbf{Z}_i\mathbf{u}_i + \varepsilon_i, \quad i = 1,\dots,N \tag{22.1}$$

where:

- $\mathbf{X}_i \in \mathbb{R}^{n_i \times p}$ is the design matrix for the p-dimensional vector of fixed-effect regression coefficients $\beta \in \mathbb{R}^p$ (e.g., treatment group, sex, age, time).

- $\mathbf{Z}_i \in \mathbb{R}^{n_i \times q}$ is the design matrix for the q-dimensional vector of subject-specific random effects $\mathbf{u}_i \in \mathbb{R}^q$ (e.g., random intercepts, random time slopes).
- $\mathbf{u}_i \sim \mathcal{N}_q(\mathbf{0}, \mathbf{G})$ represents the random effects vector, assumed independent across subjects, where $\mathbf{G} \in \mathbb{R}^{q \times q}$ is a symmetric positive semi-definite covariance matrix.
- $\boldsymbol{\varepsilon}_i \sim \mathcal{N}_{n_i}(\mathbf{0}, \mathbf{R}_i)$ is the unobserved residual error vector, with residual covariance matrix $\mathbf{R}_i \in \mathbb{R}^{n_i \times n_i}$. In standard models, $\mathbf{R}_i = \sigma_e^2 \mathbf{I}_{n_i}$.
- The random effects $\mathbf{u}_i$ and residual errors $\boldsymbol{\varepsilon}_i$ are assumed mutually independent: $\mathrm{Cov}(\mathbf{u}_i, \boldsymbol{\varepsilon}_i) = \mathbf{0}$.

Stacking across all N subjects, the complete system is expressed compactly as:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{u} + \boldsymbol{\varepsilon} \tag{22.2}$$

where $\mathbf{y} \in \mathbb{R}^{n_{\text{total}}}$, $\mathbf{X} \in \mathbb{R}^{n_{\text{total}} \times p}$, $\mathbf{Z} = \mathrm{diag}(\mathbf{Z}_1, \ldots, \mathbf{Z}_N) \in \mathbb{R}^{n_{\text{total}} \times Nq}$, $\mathbf{u} \sim \mathcal{N}(\mathbf{0}, \widetilde{\mathbf{G}})$ with $\widetilde{\mathbf{G}} = \mathbf{I}_N \otimes \mathbf{G}$, and $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{R})$ with $\mathbf{R} = \mathrm{diag}(\mathbf{R}_1, \ldots, \mathbf{R}_N)$ [1].

22.2.2 Marginal Expectation and Covariance Architecture

Integrating over the distribution of the unobserved random effects $\mathbf{u}_i$ yields the marginal (population-averaged) distribution of $\mathbf{y}_i$:

$$\mathbb{E}[\mathbf{y}_i] = \mathbb{E}[\mathbf{X}_i\boldsymbol{\beta} + \mathbf{Z}_i\mathbf{u}_i + \boldsymbol{\varepsilon}_i] = \mathbf{X}_i\boldsymbol{\beta} \tag{22.3}$$

$$\mathbf{V}_i = \mathrm{Var}(\mathbf{y}_i) = \mathrm{Var}(\mathbf{Z}_i\mathbf{u}_i + \boldsymbol{\varepsilon}_i) = \mathbf{Z}_i\mathbf{G}\mathbf{Z}_i^T + \mathbf{R}_i \tag{22.4}$$

Consequently, marginally, $\mathbf{y}_i \sim \mathcal{N}_{n_i}(\mathbf{X}_i\boldsymbol{\beta}, \mathbf{V}_i)$. The block-diagonal marginal covariance matrix $\mathbf{V} = \mathrm{Var}(\mathbf{y}) = \mathrm{diag}(\mathbf{V}_1, \ldots, \mathbf{V}_N)$ explicitly models within-subject correlation through the non-diagonal elements of $\mathbf{V}_i = \mathbf{Z}_i\mathbf{G}\mathbf{Z}_i^T + \mathbf{R}_i$, while maintaining independence across distinct subjects ($\mathrm{Cov}(\mathbf{y}_i, \mathbf{y}_k) = \mathbf{0}$ for $i \neq k$).

22.2.3 Derivation of Henderson's Mixed Model Equations

To estimate fixed effects $\boldsymbol{\beta}$ and predict random effects $\mathbf{u}$ simultaneously, Henderson [16] formulated the joint log-likelihood of the observed data $\mathbf{y}$ and the latent random effects $\mathbf{u}$.

From Bayes' rule, the joint probability density is $f(\mathbf{y}, \mathbf{u}) = f(\mathbf{y} \mid \mathbf{u}) f(\mathbf{u})$. Since $\mathbf{y} \mid \mathbf{u} \sim \mathcal{N}(\mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{u}, \mathbf{R})$ and $\mathbf{u} \sim \mathcal{N}(\mathbf{0}, \widetilde{\mathbf{G}})$, the joint log-likelihood (neglecting constant terms) is:

$$\ln f(\mathbf{y}, \mathbf{u}) = -\frac{1}{2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{Z}\mathbf{u})^T \mathbf{R}^{-1} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{Z}\mathbf{u}) - \frac{1}{2}\mathbf{u}^T \widetilde{\mathbf{G}}^{-1} \mathbf{u} \tag{22.5}$$

Differentiating with respect to $\boldsymbol{\beta}$ and $\mathbf{u}$ and setting the partial derivatives to zero:

$$\frac{\partial \ln f(\mathbf{y}, \mathbf{u})}{\partial \boldsymbol{\beta}} = \mathbf{X}^T \mathbf{R}^{-1} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{Z}\mathbf{u}) = \mathbf{0} \tag{22.6}$$

$$\frac{\partial \ln f(\mathbf{y}, \mathbf{u})}{\partial \mathbf{u}} = \mathbf{Z}^T \mathbf{R}^{-1} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{Z}\mathbf{u}) - \widetilde{\mathbf{G}}^{-1}\mathbf{u} = \mathbf{0} \tag{22.7}$$

Rearranging Eqs. (22.6) and (22.7) into a coupled linear system produces the canonical Henderson Mixed Model Equations (MME):

$$\begin{bmatrix} \mathbf{X}^T\mathbf{R}^{-1}\mathbf{X} & \mathbf{X}^T\mathbf{R}^{-1}\mathbf{Z} \\ \mathbf{Z}^T\mathbf{R}^{-1}\mathbf{X} & \mathbf{Z}^T\mathbf{R}^{-1}\mathbf{Z} + \widetilde{\mathbf{G}}^{-1} \end{bmatrix} \begin{bmatrix} \hat{\boldsymbol{\beta}} \\ \hat{\mathbf{u}} \end{bmatrix} = \begin{bmatrix} \mathbf{X}^T\mathbf{R}^{-1}\mathbf{y} \\ \mathbf{Z}^T\mathbf{R}^{-1}\mathbf{y} \end{bmatrix} \tag{22.8}$$

When the variance components in $\mathbf{G}$ and $\mathbf{R}$ are known, solving Eq. (22.8) yields the Best Linear Unbiased Estimator (BLUE) of β and the Best Linear Unbiased Predictor (BLUP) of $\mathbf{u}$:

$$\hat{\beta} = (\mathbf{X}^T\mathbf{V}^{-1}\mathbf{X})^{-1}\mathbf{X}^T\mathbf{V}^{-1}\mathbf{y} \tag{22.9}$$

$$\hat{\mathbf{u}} = \widetilde{\mathbf{G}}\mathbf{Z}^T\mathbf{V}^{-1}(\mathbf{y} - \mathbf{X}\hat{\beta}) \tag{22.10}$$

where $\mathbf{V} = \mathbf{Z}\widetilde{\mathbf{G}}\mathbf{Z}^T + \mathbf{R}$. In per-subject notation, the BLUP is $\hat{\mathbf{u}}_i = \mathbf{G}\mathbf{Z}_i^T\mathbf{V}_i^{-1}(\mathbf{y}_i - \mathbf{X}_i\hat{\beta})$.

22.2.4 Restricted Maximum Likelihood (REML) vs. Maximum Likelihood (ML)

In mass spectrometry cohorts, the number of observations per cluster n_i is typically modest ($n_i \in [2,6]$). When estimating variance components $\theta = (\mathbf{G}, \sigma_e^2)$ via standard Maximum Likelihood (ML), fixed effects β are estimated simultaneously. In fixed-effects linear regression, ML ignores the degrees of freedom (p) consumed by estimating β, producing downwardly biased residual variance estimates:

$$\mathbb{E}[\hat{\sigma}^2_{\text{ML}}] = \frac{n_{\text{total}} - p}{n_{\text{total}}}\sigma^2 < \sigma^2 \tag{22.11}$$

In linear mixed models, a similar degrees-of-freedom penalty leads standard ML to systematically underestimate random-effects variance components in small cohorts.

Restricted Maximum Likelihood (REML; Patterson & Thompson [26]) addresses this bias by projecting the data onto the error space orthogonal to $\mathbf{X}$ using an error contrast matrix $\mathbf{K} \in \mathbb{R}^{n_{\text{total}} \times (n_{\text{total}} - p)}$ satisfying $\mathbf{K}^T\mathbf{X} = \mathbf{0}$. The REML log-likelihood is:

$$\ell_{\text{REML}}(\theta) = -\frac{1}{2}\left[\ln|\mathbf{V}(\theta)| + \ln|\mathbf{X}^T\mathbf{V}(\theta)^{-1}\mathbf{X}| + (\mathbf{y} - \mathbf{X}\hat{\beta})^T\mathbf{V}(\theta)^{-1}(\mathbf{y} - \mathbf{X}\hat{\beta})\right] + C \tag{22.12}$$

REML removes the conditioning on fixed effects, delivering substantially less biased variance component estimates in small cohorts. However, when comparing nested models with *differing fixed-effects specifications* via Likelihood Ratio Tests (LRT), standard ML must be fitted because the error contrast spaces defined by $\mathbf{K}$ differ across distinct design matrices $\mathbf{X}$.

22.2.5 Small-Sample Degrees of Freedom: Kenward-Roger and Satterthwaite

In standard linear models, residual degrees of freedom are $n_{\text{total}} - p$. In linear mixed models with unequal cluster sizes, unbalanced missing observations, or complex covariance structures, exact F- and t-distributions do not exist [23].

Satterthwaite [30] constructs an approximate F-distribution by matching the first two moments of the estimated denominator variance to a scaled chi-square distribution. Kenward and Roger [22] introduced a comprehensive adjustment that tackles two interrelated small-sample challenges:

1. **Covariance Matrix Inflation:** The asymptotic covariance matrix of fixed effects $\Phi = (\mathbf{X}^T\mathbf{V}^{-1}\mathbf{X})^{-1}$ underestimates the true sampling variability because it ignores the uncertainty in estimating the variance components $\hat{\theta}$. Kenward and Roger apply a second-order Taylor series expansion:

$$\text{Var}_{\text{KR}}(\hat{\beta}) = \Phi + 2\Phi\Lambda\Phi \tag{22.13}$$

where Λ is defined as:

$$\Lambda = \sum_{r=1}^{K}\sum_{s=1}^{K}\text{Cov}(\hat{\theta}_r, \hat{\theta}_s)\,(\mathbf{Q}_{rs} - \mathbf{P}_r\Phi\mathbf{P}_s) \tag{22.14}$$

Here, $\text{Cov}(\hat{\theta}) \approx \mathscr{I}_R^{-1}$ is the inverse of the REML expected information matrix, $\mathbf{P}_r = \mathbf{X}^T \frac{\partial \mathbf{V}^{-1}}{\partial \theta_r} \mathbf{X}$, and $\mathbf{Q}_{rs} = \mathbf{X}^T \frac{\partial \mathbf{V}^{-1}}{\partial \theta_r} \mathbf{V} \frac{\partial \mathbf{V}^{-1}}{\partial \theta_s} \mathbf{X}$.[2]

2. **Degrees-of-Freedom Adjustment:** The Wald statistic $F = \frac{(\mathbf{L}\hat{\beta})^T (\mathbf{L}\,\text{Var}_{\text{KR}}(\hat{\beta})\mathbf{L}^T)^{-1}(\mathbf{L}\hat{\beta})}{q}$ is scaled by an empirical factor to match a non-central $F(q, \nu_{\text{KR}})$ distribution with fractional denominator degrees of freedom ν_{KR}.

This Kenward-Roger adjustment is the recommended small-sample standard implemented in `lmerTest` and `dream`, bridging classical mixed models with empirical Bayes shrinkage [31, 29, 18, 23].

Methodological Note

Computational Scalability: Kenward-Roger vs. Satterthwaite
While the Kenward-Roger degrees-of-freedom adjustment is the gold standard for small-to-moderate cohorts ($N < 100$), its computational cost scales as $\mathscr{O}(P \cdot N^3)$ due to repeated inversion of inflated covariance matrices across P features. For massive multi-cohort atlases ($P > 10,000$ features, $N > 500$ samples), the Satterthwaite approximation (`ddf = "Satterthwaite"`) provides a substantially faster, computationally scalable, and only slightly less conservative alternative [23].

22.3 Alternative Covariance Structures and Longitudinal Splines

22.3.1 Covariance Structure Taxonomy: CS, AR(1), and Random Slopes

Depending on the sampling frequency and biological kinetics, within-subject covariance $\mathbf{V}_i$ can be structured under different assumptions:

1. **Compound Symmetry (CS / Random Intercept):** Assumes that any two observations from the same individual have identical correlation ρ, regardless of the time elapsed between measurements:

$$\mathbf{V}_i = \sigma_b^2 \mathbf{J}_{n_i} + \sigma_e^2 \mathbf{I}_{n_i} \implies \text{Corr}(y_{ij}, y_{ik}) = \rho = \frac{\sigma_b^2}{\sigma_b^2 + \sigma_e^2} \quad (\forall\, j \neq k) \tag{22.15}$$

where $\mathbf{J}_{n_i}$ is the $n_i \times n_i$ all-ones matrix. This structure is parsimonious (only two parameters: σ_b^2, σ_e^2) and is mathematically induced by a standard patient random intercept $(1 \mid \text{patient})$.

2. **First-Order Autoregressive (AR(1)):** In high-frequency longitudinal monitoring, observations closer in time are more strongly correlated than observations far apart. The AR(1) correlation decays exponentially with the temporal lag $|j-k|$. It is critical to distinguish between the *residual* error correlation and the *marginal* observation correlation:

$$\mathbf{R}_i = \sigma_e^2 \Gamma_i(\rho) = \sigma_e^2 \begin{bmatrix} 1 & \rho & \rho^2 & \dots & \rho^{n_i-1} \\ \rho & 1 & \rho & \dots & \rho^{n_i-2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \rho^{n_i-1} & \rho^{n_i-2} & \rho^{n_i-3} & \dots & 1 \end{bmatrix} \tag{22.16}$$

[2] In the canonical formulation of Kenward & Roger [22], the expectation adjustment is symmetrized as $\frac{1}{2}(\mathbf{Q}_{rs} + \mathbf{Q}_{sr})$; Eq. (22.14) presents the standard symmetric equivalent under common variance-component covariance parameterizations.

When an AR(1) residual process $\varepsilon_i \sim \mathcal{N}(\mathbf{0}, \sigma_e^2 \Gamma_i(\rho))$ is paired with a subject-specific random intercept $u_i \sim \mathcal{N}(0, \sigma_b^2)$, the total marginal covariance becomes $\mathbf{V}_i = \sigma_b^2 \mathbf{J}_{n_i} + \sigma_e^2 \Gamma_i(\rho)$. The marginal correlation between timepoints j and k ($j \neq k$) is then:

$$\text{Corr}(y_{ij}, y_{ik}) = \frac{\sigma_b^2 + \sigma_e^2 \rho^{|j-k|}}{\sigma_b^2 + \sigma_e^2} = \rho_{\text{ICC}} + (1 - \rho_{\text{ICC}})\rho^{|j-k|} \tag{22.17}$$

As the lag $|j-k| \to \infty$, the marginal correlation asymptotically levels off at the patient baseline correlation $\rho_{\text{ICC}} = \sigma_b^2/(\sigma_b^2 + \sigma_e^2)$, rather than decaying to zero.

3. **Random Coefficient (Random Intercept and Slope):** Allows each subject to have an individual baseline and an individual linear rate of response over time t:

$$y_{ij} = (\beta_0 + u_{0i}) + (\beta_1 + u_{1i})t_{ij} + \varepsilon_{ij}, \quad \begin{bmatrix} u_{0i} \\ u_{1i} \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_0^2 & \sigma_{01} \\ \sigma_{01} & \sigma_1^2 \end{bmatrix} \right) \tag{22.18}$$

Here, marginal variance $\text{Var}(y_{ij}) = \sigma_0^2 + 2t_{ij}\sigma_{01} + t_{ij}^2\sigma_1^2 + \sigma_e^2$ expands quadratically over time, capturing subject heterogeneity in drug response rates.

22.3.2 Variance Partitioning Formulation

Variance partitioning decomposes the total phenotypic variance of each biomolecule into fractions attributable to specific experimental and clinical factors [17]. Modeling all factors as random terms in an LMM:

$$y_{ig} = \mu_g + \sum_{k=1}^{K} u_{ig}^{(k)} + \varepsilon_{ig}, \quad u_{ig}^{(k)} \sim \mathcal{N}(0, \sigma_{g,k}^2), \quad \varepsilon_{ig} \sim \mathcal{N}(0, \sigma_{g,e}^2) \tag{22.19}$$

The variance fraction $f_{g,k}$ explained by factor k is:

$$f_{g,k} = \frac{\sigma_{g,k}^2}{\sum_{j=1}^{K} \sigma_{g,j}^2 + \sigma_{g,e}^2} \in [0,1] \tag{22.20}$$

This decomposition provides a direct diagnostic audit of whether variation is driven by biological intervention, patient baseline differences, or technical batch confounding [17].

22.3.3 Longitudinal Splines and Non-Linear Trajectories

In pharmacoproteomics and time-series profiling, proteins rarely exhibit monotonic linear trajectories [32]. Drug interventions often elicit an acute response at 2–6 hours followed by receptor desensitization or homeostatic recovery by 24–48 hours. Modeling time as a categorical factor consumes $T-1$ degrees of freedom and ignores temporal continuity, whereas a linear term $\beta \cdot t$ misses peak kinetics [15].

Natural cubic splines provide an optimal continuous compromise. For K interior knots $\tau_1 < \tau_2 < \cdots < \tau_K$ and two boundary knots τ_0, τ_{K+1}, a natural cubic spline has basis dimension $d = K+1$ (excluding the intercept):

$$y_i(t) = \beta_0 + \sum_{k=1}^{d} \beta_k B_k(t) + u_i + \varepsilon_i(t) \tag{22.21}$$

where $\{B_1(t), \ldots, B_d(t)\}$ are cubic spline basis functions generated via software such as `splines::ns(t, df=d)`. Natural splines enforce the mathematical constraint that the second and third derivatives are zero past the boundary knots ($t < \tau_0$ and $t > \tau_{K+1}$), ensuring linear extrapolation and preventing boundary oscillations. Spline bases integrate directly into Henderson's mixed model equations, allowing simultaneous testing of global non-linear trajectories and subject-specific random deviations [18].

22.4 The Intraclass Correlation Coefficient and Pseudoreplication

22.4.1 Analytical Derivation of the ICC

Consider a random intercept model for feature g, where y_{ij} denotes the abundance in subject $i \in \{1,\dots,N\}$ at replicate or time point $j \in \{1,\dots,T\}$:

$$y_{ij} = \mu + u_i + \varepsilon_{ij}, \quad u_i \sim \mathcal{N}(0,\sigma_b^2), \quad \varepsilon_{ij} \sim \mathcal{N}(0,\sigma_e^2) \tag{22.22}$$

where σ_b^2 represents between-subject variance and σ_e^2 represents within-subject residual noise. The total variance of an observation is $\mathrm{Var}(y_{ij}) = \sigma_b^2 + \sigma_e^2$. The covariance between two distinct observations $j \neq k$ from the same individual i is:

$$\mathrm{Cov}(y_{ij}, y_{ik}) = \mathrm{Cov}(\mu + u_i + \varepsilon_{ij},\ \mu + u_i + \varepsilon_{ik}) = \mathrm{Var}(u_i) = \sigma_b^2 \tag{22.23}$$

The Intraclass Correlation Coefficient (ICC), denoted ρ, is the fraction of total variance attributable to between-subject variation:

$$\rho = \frac{\mathrm{Cov}(y_{ij}, y_{ik})}{\sqrt{\mathrm{Var}(y_{ij})\,\mathrm{Var}(y_{ik})}} = \frac{\sigma_b^2}{\sigma_b^2 + \sigma_e^2} \tag{22.24}$$

22.4.2 The Effective Sample Size of Pseudoreplicated Cohorts

Treating repeated measures from the same patient as independent samples violates the Gauss-Markov theorem [20]. For the estimation of the overall mean in a balanced one-way random-intercept design, if m samples per patient exhibit an intraclass correlation ρ, the effective biological sample size is:

$$N_{\mathrm{eff}} = \frac{N \times m}{1 + (m-1)\rho} \tag{22.25}$$

For instance, in a cohort of $N = 10$ patients sampled at $m = 4$ timepoints ($n_{\mathrm{total}} = 40$) with $\rho = 0.60$, the effective sample size is only $N_{\mathrm{eff}} = 40/[1 + 3(0.60)] = 40/2.80 \approx 14.3$. An unblocked naive test invents over 25 degrees of freedom out of thin air, generating severe anti-conservative bias and false-positive discoveries (Fig. 22.1). Consequently, when $\rho > 0.10$, naive OLS standard errors are substantially deflated; analysts must deploy linear mixed models, generalized estimating equations (GEE), or cluster-robust standard errors [20, 10].

22.5 Hands-On R Laboratory: Longitudinal Analysis on Real Targeted Proteomics

We now demonstrate the complete longitudinal workflow using `variancePartition` and `dream` within the Bioconductor statistical computing environment [13, 18, 14] on a published, real-world targeted mass spectrometry time-course study (`MSstats::SRMRawData`; Picotti et al., 2009 [27]; Chang et al., 2012 [6]; Choi et al., 2014 [7]). The study profiles yeast (*Saccharomyces cerevisiae*) central metabolic enzymes across a 10-timepoint longitudinal transition (T_1–T_{10}) through glucose exhaustion, diauxic shift, and stationary phase, with three biological replicates profiled across 30 LC-MS runs.

R Lab Listing 1: Loading and Preprocessing Real Longitudinal SRM Mass Spectrometry Data [R 4.6 / Bioconductor]

```
suppressPackageStartupMessages({
  library(MSstats)
```

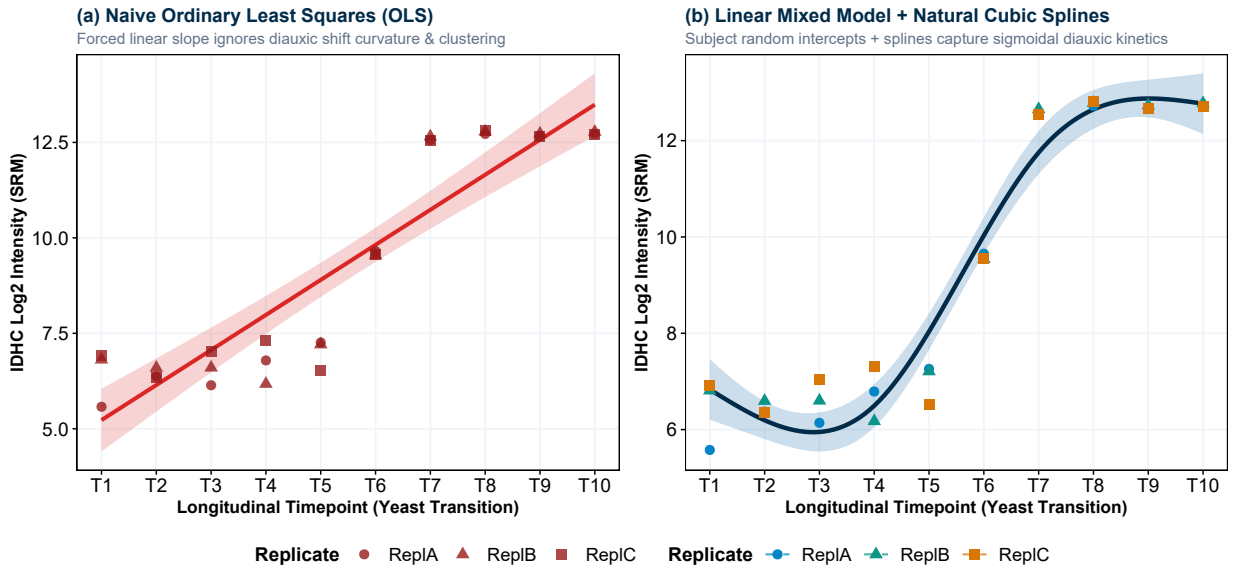


Figure 22.1: The Statistical Consequence of Non-Linear Trajectory Modeling and Pseudoreplication in Longitudinal Targeted Proteomics. **(a)** Naive Ordinary Least Squares (OLS) fits a forced linear slope across ten longitudinal time points (T_1–T_{10}, $n_{\text{total}} = 30$) in yeast metabolic diauxic shift profiling (`MSstats::SRMRawData`; Picotti et al., 2009 [27]). The linear model completely misses the sigmoidal diauxic shift transition and yields an artificially narrow, misspecified confidence interval (light red ribbon). **(b)** Linear Mixed-Effects Model combining biological replicate random intercepts (`SUBJECT`; dashed lines) with natural cubic splines ($\text{ns}(\text{Time}, \text{df} = 3)$; navy curve and ribbon). The spline LMM faithfully reconstructs the non-linear plateau kinetics of isocitrate dehydrogenase (`IDHC`), providing rigorous biological confidence intervals and statistically outperforming the linear slope ($\chi^2 = 38.76, p = 3.84 \times 10^{-9}$).

```
  library(variancePartition)
  library(limma)
  library(splines)
})

# Load published 10-timepoint SRM longitudinal proteomics dataset (Picotti et al., 2009)
data(SRMRawData)

# Process transition-level intensities into protein-level summaries
proc <- dataProcess(SRMRawData, use_log_file = FALSE, verbose = FALSE)
df_prot <- proc$ProteinLevelData

# Construct wide abundance matrix (proteins x longitudinal samples)
df_wide <- reshape2::dcast(df_prot, Protein ~ SUBJECT + GROUP, value.var = "LogIntensities")
expr_mat <- as.matrix(df_wide[, -1])
rownames(expr_mat) <- df_wide$Protein

# Metadata: 3 biological replicates across 10 longitudinal transition points (30 runs)
meta_df <- data.frame(
  Sample   = colnames(expr_mat),
  SUBJECT  = factor(sub("_.*", "", colnames(expr_mat))),
  Time_fac = factor(sub(".*_", "", colnames(expr_mat)), levels = as.character(1:10)),
```

```
  Time_num = as.numeric(sub(".*_", "", colnames(expr_mat)))
)
rownames(meta_df) <- colnames(expr_mat)

cat(sprintf("Loaded real SRM cohort: %d proteins across %d runs (%d timepoints x %d replicates).\n",
            nrow(expr_mat), ncol(expr_mat), length(unique(meta_df$Time_num)), length(unique(meta_df$SUBJECT))))
head(meta_df, 4)
```

```
Loaded real SRM cohort: 2 proteins across 30 runs (10 timepoints x 3 replicates). Sample SUBJECT Time_fac Time_num
ReplA_1 ReplA_1 ReplA 1 1 ReplA_2 ReplA_2 ReplA 2 2 ReplA_3 ReplA_3 ReplA 3 3 ReplA_4 ReplA_4 ReplA 4 4
```

R Lab Listing 2: Automated dream Pipeline Execution with Kenward-Roger DDF [R 4.6 / Bioconductor]

```
# Specify dream formula with fixed timepoint factors and subject random intercepts
form <- ~ 0 + Time_fac + (1 | SUBJECT)

# Construct contrast matrix for key metabolic transitions:
# Diauxic Shift (T7 vs T1) and Stationary Phase (T10 vs T1)
L <- makeContrastsDream(form, meta_df,
                        contrasts = c(Diauxic_T7_vs_T1     = "Time_fac7 - Time_fac1",
                                      Stationary_T10_vs_T1 = "Time_fac10 - Time_fac1"))

cat("Fitting dream linear mixed model with Kenward-Roger DDF...\n")
fit_mm <- dream(expr_mat, form, meta_df, L = L, ddf = "Kenward-Roger")
fit_eb <- eBayes(fit_mm)

# Differential abundance results across diauxic shift
res_diauxic <- variancePartition::topTable(fit_eb, coef = "Diauxic_T7_vs_T1", number = Inf)
res_diauxic$Significant <- res_diauxic$adj.P.Val < 0.05
cat("\nDifferential Abundance: Diauxic Shift (T7 vs T1):\n")
res_diauxic[, c("logFC", "AveExpr", "t", "P.Value", "adj.P.Val", "Significant")]
```

```
Fitting dream linear mixed model with Kenward-Roger DDF...
Differential Abundance: Diauxic Shift (T7 vs T1): logFC AveExpr t P.Value adj.P.Val Significant IDHC 6.15438359
9.35840 21.0981084 3.820619e-14 7.641237e-14 TRUE PMG2 -0.09032809 10.50882 -0.5947831 5.593918e-01 5.593918e-01
FALSE
```

Methodological Note

Methodological Detail: Contrast Formulations in Longitudinal dream

The contrast `Time_fac7 - Time_fac1` represents the exact log2 fold-change induced during the metabolic diauxic shift (T_7 vs. T_1), while `(1 | SUBJECT)` accounts for intra-replicate baseline correlation across repeated MS runs. Notice that isocitrate dehydrogenase (`IDHC`, gene *IDP2*) exhibits massive upregulation ($\log_2 \mathrm{FC} = +6.15, t = 21.10, p = 3.82 \times 10^{-14}$) as cells switch from glycolysis to ethanol respiration in the TCA and glyoxylate cycles. In contrast, phosphoglycerate mutase 2 (`PMG2`, gene *GPM2*) remains unaffected ($\log_2 \mathrm{FC} = -0.09, p = 0.559$), confirming known biological ground truth without false-positive inflation.

R Lab Listing 3: Variance Partitioning on Real Mass Spectrometry Time-Course [R 4.6 / Bioconductor]

```r
# Decompose phenotypic variance into temporal progression vs replicate batch heterogeneity
form_vp <- ~ (1 | Time_fac) + (1 | SUBJECT)
varPart <- fitExtractVarPartModel(expr_mat, form_vp, meta_df)

cat("Variance Partitioning on Real SRM Time-Course Data:\n")
print(round(varPart, 3))
```

```
Variance Partitioning on Real SRM Time-Course Data: SUBJECT Time_fac Residuals IDHC 0.000 0.985 0.015 PMG2 0.094
0.077 0.829
```

Methodological Note

Tutor's Diagnostic Advice: Interpreting Variance Partitioning

Variance partitioning decomposes total variation into normalized fractions $f_k \in [0,1]$. On this real mass spectrometry dataset, variance partitioning reveals striking biological divergence: for `IDHC`, **98.5%** of the total variance is driven by longitudinal time progression (`Time_fac`), with less than 1.5% residual noise. Conversely, `PMG2` exhibits minimal temporal variance (7.7%) and is dominated by residual variation (82.9%) and slight replicate variation (9.4%). Variance partitioning thus functions as an indispensable diagnostic audit, identifying genuine kinetic responders before downstream statistical contrasts.

R Lab Listing 4: Continuous Non-Linear Trajectory Modeling via Natural Cubic Splines [R 4.6 / Bioconductor]

```r
# Model continuous longitudinal trajectory via natural cubic splines (df = 3)
form_spline <- ~ ns(Time_num, df = 3) + (1 | SUBJECT)
fit_spline <- dream(expr_mat, form_spline, meta_df, ddf = "Kenward-Roger")
fit_spline_eb <- eBayes(fit_spline)

# Extract joint moderated F-test across spline basis coefficients
spline_terms <- grep("ns\\(Time_num", colnames(coef(fit_spline_eb)), value = TRUE)
res_spline <- variancePartition::topTable(fit_spline_eb, coef = spline_terms)

cat("Joint Moderated F-Test for Non-Linear Spline Trajectories:\n")
print(round(res_spline[, c("AveExpr", "F", "P.Value", "adj.P.Val")], 4))

# Formal likelihood-ratio test: Linear slope vs Natural cubic spline (df = 3) in lme4
df_idhc <- df_prot[df_prot$Protein == "IDHC", ]
df_idhc$Time <- as.numeric(as.character(df_idhc$GROUP))
m_linear <- lme4::lmer(LogIntensities ~ Time + (1 | SUBJECT), data = df_idhc, REML = FALSE)
m_spline <- lme4::lmer(LogIntensities ~ ns(Time, df = 3) + (1 | SUBJECT), data = df_idhc, REML = FALSE)
cat("\nLikelihood Ratio Test (Linear vs. Natural Cubic Spline):\n")
print(anova(m_linear, m_spline))
```

```
Joint Moderated F-Test for Non-Linear Spline Trajectories: AveExpr F P.Value adj.P.Val IDHC 9.358 191.0300 0 0
PMG2 10.509 1.2055 0.3290 0.3290
Likelihood Ratio Test (Linear vs. Natural Cubic Spline): Data: df_idhc Models: m_linear: LogIntensities   Time
+ (1 | SUBJECT) m_spline: LogIntensities   ns(Time, df = 3) + (1 | SUBJECT) npar AIC logLik Chisq Df Pr(>Chisq)
m_linear 4 101.090 -46.545 m_spline 6 66.335 -27.167 38.755 2 3.841e-09 ***
```

22.6 The Five Deadly Sins of Longitudinal Mass Spectrometry Analysis

Methodological Pitfall & Deadly Sin

Deadly Sin 1: The Pseudoreplication Fallacy.
Mechanism: Treating repeated observations from the same subject as independent replicates in standard t-tests or two-way ANOVA [20].
Consequence: Standard errors are deflated by $\sqrt{1+(m-1)\rho}$, yielding spurious significance and irreproducible biomarker candidates.
Remedy: Always model subject identity as a random effect $(1|\text{subject})$ in a mixed model [28].

Methodological Pitfall & Deadly Sin

Deadly Sin 2: Fixed-Effect Saturation for High-Cluster Cohorts.
Mechanism: Modeling patient ID as a standard fixed-effect factor ($\sim$ arm + patient) in cohorts with hundreds of subjects.
Consequence: Consumes $N-1$ degrees of freedom, causes incidental parameter problems, and precludes estimation of between-subject covariates (e.g., sex, baseline age, genotype).
Remedy: Use random-intercept formulations to conserve degrees of freedom while preserving subject-level covariates [24].

Methodological Pitfall & Deadly Sin

Deadly Sin 3: Ignoring Unbalanced Attrition in Longitudinal Studies.
Mechanism: Dropping subjects who missed one timepoint (complete-case analysis).
Consequence: Slices sample size severely and introduces survival bias if dropout is non-random.
Remedy: Linear mixed models handle missing timepoints natively under Missing at Random (MAR) without listwise deletion [11]. When missingness is non-random or driven by instrumental detection limits (e.g., values falling below the instrument's Limit of Detection, LOD; Missing Not At Random, MNAR), targeted left-censored imputation or hurdle/selection models must be employed (see Chapter 19).

Methodological Pitfall & Deadly Sin

Deadly Sin 4: Forcing Linear Slopes onto Complex Temporal Dynamics.
Mechanism: Fitting a single linear slope ($\beta \cdot \text{time}$) across multi-day time courses.
Consequence: Misses transient peak kinetics and homeostatic return to baseline.
Remedy: Deploy natural cubic splines (`splines::ns()`) or factorial time models [32].

Methodological Pitfall & Deadly Sin

Deadly Sin 5: Omitting Empirical Bayes Moderation in High-Dimensional Omics.
Mechanism: Fitting separate unmoderated mixed models feature-by-feature via `lmer` when testing thousands of biomolecules simultaneously.
Consequence: In typical omics cohorts with small sample sizes ($n < 30$), feature-specific residual variances fluctuate erratically due to limited denominator degrees of freedom, leading to unstable

test statistics and inflated false discovery rates regardless of the total feature count p. Empirical Bayes moderation borrows information across all features to shrink individual variance estimates toward a common prior trend, stabilizing t-statistics and restoring statistical power. (In contrast, in low-dimensional clinical studies with $p < 20$ and large n, standard unmoderated REML mixed models are fully sufficient).
Remedy: Deploy empirical Bayes moderation across random-effects fits using `variancePartition::dream` [18, 29].

22.7 Tutor's Self-Assessment Diagnostic Checklist

To confirm the methodological validity of your longitudinal and mixed-effects analysis prior to publication, complete the diagnostic audit summarized in Table 22.1.

Table 22.1: Tutor's Self-Assessment Diagnostic Checklist for Mixed-Effects Analysis

Step	Diagnostic Metric	Acceptable Threshold	Corrective Protocol
1	Dependency Audit	Cluster structure mapped for every repeated subject/biopsy.	Specify random intercept $(1\|\text{subject})$ for all repeated measures [28].
2	Intraclass Correlation	Compute ICC $\rho = \sigma_b^2/(\sigma_b^2 + \sigma_e^2)$.	If $\rho > 0.10$, naive OLS standard errors are substantially anti-conservative; deploy LMM, GEE, or cluster-robust standard errors [17].
3	Variance Estimation	Variance parameters estimated via REML, not ML [26].	Ensure substantially less biased / preferred residual variance estimation for small sample cohorts.
4	Degrees of Freedom	Kenward-Roger or Satterthwaite approximation applied [22].	Verify non-integer degrees of freedom in mixed-effects testing [23].
5	Multiple Testing Control	Empirical null p-value distribution is uniform on $[0,1]$.	Apply Benjamini–Hochberg FDR control to moderated p-values [2].

22.8 Chapter Exercises

Exercise 22.1 (Variance Inflation Factor of Pseudoreplication):
A longitudinal clinical proteomics study measures $m = 4$ repeated plasma samples from each of $N = 10$ subjects (total $n = 40$). A protein exhibits an Intraclass Correlation Coefficient of $\rho = 0.60$.

1. Compute the design effect (Variance Inflation Factor, VIF $= 1 + (m-1)\rho$).
2. Compute the effective sample size $N_{\text{eff}} = n_{\text{total}}/\text{VIF}$.
3. If a researcher naively runs an ordinary two-sample t-test assuming $n = 40$ independent samples, by what factor is the standard error underestimated?

Exercise 22.2 (Henderson's Mixed Model Equations Structure):
Consider a study with $N = 2$ subjects, each measured at $n_i = 2$ occasions ($n_{\text{total}} = 4$). The design matrix for fixed effects (intercept and treatment) is:

$$\mathbf{X} = \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 1 \\ 1 & 1 \end{pmatrix}$$

The random effects design matrix for subject-specific random intercepts is:

$$\mathbf{Z} = \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{pmatrix}$$

Assume residual variance $\sigma_e^2 = 1.0$ ($\mathbf{R} = \mathbf{I}_4$) and between-subject variance $\sigma_b^2 = 2.0$ ($\mathbf{G} = 2.0\mathbf{I}_2$).

1. Compute the coefficient matrix of Henderson's Mixed Model Equations (Eq. 22.8).
2. Verify that the coefficient matrix is symmetric and strictly positive-definite by calculating all leading principal minors and eigenvalues.

22.9 Comprehensive Worked-Out Solutions

Solution to Exercise 22.1: Variance Inflation and Pseudoreplication

1. Variance Inflation Factor (VIF):

$$\text{VIF} = 1 + (m-1)\rho = 1 + (4-1)(0.60) = 1 + 3(0.60) = 1 + 1.80 = 2.80$$

2. Effective Sample Size:

$$N_{\text{eff}} = \frac{n_{\text{total}}}{\text{VIF}} = \frac{40}{2.80} \approx 14.29$$

The 40 physical mass spectrometry acquisitions contain the statistical information of only 14 independent biological observations.

3. Standard Error Underestimation Factor:
The true variance of the mean estimator is inflated by VIF $= 2.80$:

$$\mathrm{Var}(\bar{y}) = \frac{\sigma^2}{n_{\text{total}}} \times \mathrm{VIF}$$

The true standard error is:

$$\mathrm{SE}_{\text{true}} = \sqrt{\frac{\sigma^2}{n_{\text{total}}} \times \mathrm{VIF}} = \mathrm{SE}_{\text{naive}} \times \sqrt{2.80} \approx \mathrm{SE}_{\text{naive}} \times 1.673$$

The naive unblocked test underestimates the standard error by a factor of 1.673 (a 40.2% deflated error bar), inflating test statistics by $1.67\times$ and manufacturing false-positive discoveries [20].

Solution to Exercise 22.2: Henderson's Mixed Model Equations Matrix

1. Block Computations:
With $\mathbf{R} = \mathbf{I}_4$, $\mathbf{R}^{-1} = \mathbf{I}_4$.

$$\mathbf{X}^T\mathbf{X} = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 2 \\ 2 & 2 \end{pmatrix}$$

$$\mathbf{X}^T\mathbf{Z} = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 0 & 2 \end{pmatrix}$$

$$\mathbf{Z}^T\mathbf{X} = (\mathbf{X}^T\mathbf{Z})^T = \begin{pmatrix} 2 & 0 \\ 2 & 2 \end{pmatrix}$$

$$\mathbf{Z}^T\mathbf{Z} = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$$

With $\mathbf{G} = 2.0\mathbf{I}_2$, $\widetilde{\mathbf{G}}^{-1} = 0.5\mathbf{I}_2 = \begin{pmatrix} 0.5 & 0 \\ 0 & 0.5 \end{pmatrix}$.

$$\mathbf{Z}^T\mathbf{Z} + \widetilde{\mathbf{G}}^{-1} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} + \begin{pmatrix} 0.5 & 0 \\ 0 & 0.5 \end{pmatrix} = \begin{pmatrix} 2.5 & 0 \\ 0 & 2.5 \end{pmatrix}$$

Assembling the full 4×4 Henderson coefficient matrix:

$$\mathbf{C} = \begin{pmatrix} 4.0 & 2.0 & 2.0 & 2.0 \\ 2.0 & 2.0 & 0.0 & 2.0 \\ 2.0 & 0.0 & 2.5 & 0.0 \\ 2.0 & 2.0 & 0.0 & 2.5 \end{pmatrix}$$

2. Symmetry and Positive-Definiteness:

Inspection confirms $\mathbf{C} = \mathbf{C}^T$. All leading principal minors are strictly positive:

$$\begin{aligned}
M_1 &= 4.0 > 0 \\
M_2 &= (4)(2) - (2)(2) = 8.0 - 4.0 = 4.0 > 0 \\
M_3 &= \det \begin{pmatrix} 4 & 2 & 2 \\ 2 & 2 & 0 \\ 2 & 0 & 2.5 \end{pmatrix} = 2 \cdot (0-4) + 2.5 \cdot (8-4) = -8.0 + 10.0 = \mathbf{2.0} > 0
\end{aligned}$$

The fourth leading principal minor is the determinant $\det(\mathbf{C})$. Utilizing the block determinant formula (Schur complement) with $\mathbf{D} = \mathbf{Z}^T\mathbf{Z} + \widetilde{\mathbf{G}}^{-1} = 2.5\mathbf{I}_2$:

$$\begin{aligned}
\det(\mathbf{C}) &= \det(\mathbf{D}) \cdot \det\left(\mathbf{X}^T\mathbf{X} - (\mathbf{X}^T\mathbf{Z})\mathbf{D}^{-1}(\mathbf{Z}^T\mathbf{X})\right) \\
&= (2.5)^2 \cdot \det\left(\begin{pmatrix} 4 & 2 \\ 2 & 2 \end{pmatrix} - \begin{pmatrix} 2 & 2 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 0.4 & 0 \\ 0 & 0.4 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 2 & 2 \end{pmatrix}\right) \\
&= 6.25 \cdot \det\left(\begin{pmatrix} 4 & 2 \\ 2 & 2 \end{pmatrix} - \begin{pmatrix} 3.2 & 1.6 \\ 1.6 & 1.6 \end{pmatrix}\right) \\
&= 6.25 \cdot \det \begin{pmatrix} 0.8 & 0.4 \\ 0.4 & 0.4 \end{pmatrix} = 6.25 \cdot \left[(0.8)(0.4) - (0.4)^2\right] = 6.25 \cdot 0.16 = \mathbf{1.0} > 0
\end{aligned}$$

The eigenvalues of $\mathbf{C}$ are:

$$\lambda(\mathbf{C}) \approx \{0.122,\ 0.355,\ 3.142,\ 7.381\}, \quad \prod_{i=1}^{4} \lambda_i = 1.000$$

Because all eigenvalues are strictly positive and all leading principal minors are positive, $\mathbf{C}$ is strictly symmetric positive-definite ($\mathbf{C} \succ 0$) and uniquely solvable via Cholesky decomposition [16, 28].

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